

Logistic Growth

A logistic growth function can be used to model population growth.

Logistic Growth Functions

A **logistic growth function** is a function of the form

$$f(x) = \frac{C}{1 + Ae^{-Bx}}$$

where A , B , and C are all positive constants.

Evaluating Logistic Growth Functions

EXAMPLE 1

Evaluate $f(x) = \frac{200}{1 + 4e^{-1.2x}}$.

A $f(0)$

SOLUTION

$$f(0) = \frac{200}{1 + 4e^{-1.2(0)}} = \frac{200}{1 + 4e^0} = \frac{200}{1 + 4} = 40$$

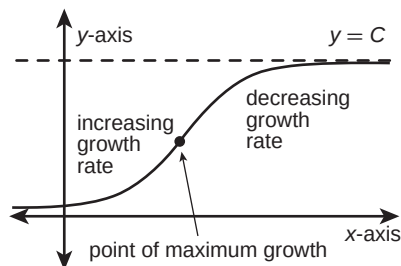
B $f(1)$

SOLUTION

$$f(1) = \frac{200}{1 + 4e^{-1.2(1)}} = \frac{200}{1 + 4e^{-1.2}} \approx 90.712 \quad \blacksquare$$

Graphing Logistic Growth Functions

Unlike exponential growth functions, where the value of $f(x)$ increases without bound as x increases, a logistic growth function $f(x) = \frac{C}{1 + Ae^{-Bx}}$ has an upper bound at $y = C$.



Graphs of Logistic Growth Functions

The graph of a logistic growth function $f(x) = \frac{C}{1 + Ae^{-Bx}}$ has the following features:

- The y-intercept is $\frac{C}{1 + A}$.
- The lines $y = 0$ and $y = C$ are horizontal asymptotes.
- The point of maximum growth is $\left(\frac{\ln A}{B}, \frac{C}{2}\right)$.
- To the left of the point of maximum growth, the rate of growth is increasing.
- To the right of the point of maximum growth, the rate of growth is decreasing.

EXAMPLE 2

For the function $f(x) = \frac{6}{1 + 2e^{-0.5x}}$:

A Identify the y -intercept.

SOLUTION

$$A = 2 \quad C = 6$$

The y -intercept is $\frac{C}{1 + A} = \frac{6}{1 + 2} = 2$.

B Identify the asymptotes.

SOLUTION

The lines $y = 0$ and $y = 6$ are asymptotes.

C Identify the point of maximum growth rate.

SOLUTION

The point of maximum growth is $\left(\frac{\ln A}{B}, \frac{C}{2}\right)$.

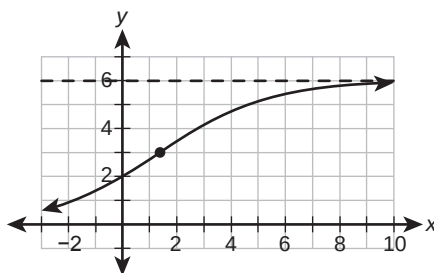
$$A = 2 \quad B = 0.5 \quad C = 6$$

The point of maximum growth rate is $\left(\frac{\ln A}{B}, \frac{C}{2}\right) = \left(\frac{\ln 2}{0.5}, \frac{6}{2}\right) \approx (1.4, 3)$.

D Make a table of values and sketch the graph.

SOLUTION

x	-2	2	4	6
$f(x)$	0.9	3.5	4.7	5.5



Solving Logistic Growth Functions

EXAMPLE 3

Given the function $f(x) = \frac{20}{1 + 2e^{-2x}}$, find the value of x when $f(x) = 8$.

SOLUTION

$$f(x) = \frac{20}{1 + 2e^{-2x}}$$

$$8 = \frac{20}{1 + 2e^{-2x}}$$

$$8(1 + 2e^{-2x}) = 20$$

$$8 + 16e^{-2x} = 20$$

$$16e^{-2x} = 12$$

$$e^{-2x} = 0.75$$

$$-2x = \ln 0.75$$

$$x = -\frac{1}{2} \ln 0.75$$

$$x \approx 0.1438$$

$x \approx 0.1438$ when $f(x) = 8$ ■